

Excerpts from Formalized Music

Iannis Xenakis

Chapter V: Free Stochastic Music by Computer

After this interlude, we return to the treatment of composition by machines.

The theory put forward by *Achorripsis* had to wait four years before being realized mechanically. This realization occurred thanks to M. François Génuys of IBM-France and to M. Jacques Barraud of the Régie Autonome des Transports Parisiens.

The Paradox: Music and Computers

A Stochastic Work Executed by the IBM-7090

The general public has a number of different reactions when faced by the alliance of the machine with artistic creation. They fall into three categories:

“It is impossible to obtain a *work of art*, since by definition it is a handicraft and requires moment-by-moment “creation” for each detail and for the entire structure, while a machine is an inert thing and cannot invent.”

“Yes, one may play games with a machine or use it for speculative purposes, but the result will not be “finished”: it will represent only an experiment—interesting, perhaps, but no more.”

The enthusiasts who at the outset accept without flinching the whole frantic brouhaha of science fiction. “The moon? Well, yes, it’s within our reach. Prolonged life will also be with us tomorrow—why not a creative machine?” These people are among the credulous who, in their, idiosyncratic optimism, have replaced the myths of Icarus and the fairies, which have decayed, by the scientific civilization of the twentieth century, and science partly agrees with them. In reality, science is neither all paradox nor all animism, for it progresses in limited stages that are not foreseeable at too great a distance.

There exists in all the arts what we may call rationalism in the etymological sense: the search for proportion. The *artist* has always called upon it out of *necessity*. The rules of construction have varied widely over the centuries, but there have always been rules in every epoch because of the necessity of making oneself understood. Those who believe the first statement above are the first to refuse to apply the qualification *artistic* to a product which they do not *understand* at all.

Thus the musical scale is a convention which circumscribes the area of potentiality and permits construction within those limits in its own particular symmetry. The rules of Christian hymnography, of harmony, and of counterpoint in the various ages have allowed artists to construct and to make themselves understood by those who adopted the same constraints—through traditions, through collective taste or imitation, or through sympathetic resonance. The rules of serialism, for instance, those that banned the traditional octave doublings of tonality, imposed constraints which were partly new but none the less real.

Now everything that is rule or repeated constraint is part of the mental machine. A little “imaginary machine,” Philippot would have said—a choice, a set of decisions. A musical work can be analyzed as a multitude of mental machines. A melodic theme in a symphony is a mold, a mental machine, in the

same way as its structure is. These mental machines are something very restrictive and deterministic, and sometimes very vague and indecisive. In the last few years we have seen that this idea of mechanism is really a very general one. It flows through every area of human knowledge and action, from strict logic to artistic manifestations.

Just as the wheel was once one of the greatest products of human intelligence, a mechanism which allowed one to travel farther and faster with more luggage, so is the computer, which today allows the transformation of man's ideas. Computers resolve logical problems by heuristic methods. But computers are not really responsible for the introduction of mathematics into music; rather it is mathematics that makes use of the computer in composition. Yet if people's minds are in general ready to recognize the usefulness of geometry in the plastic arts (architecture, painting, etc.), they have only one more stream to cross to be able to conceive of using more abstract, non-visual mathematics and machines as aids to musical composition, which is more abstract than the plastic arts.

To summarize:

1. The creative thought of man gives birth to mental mechanisms, which, in the last analysis, are merely sets of constraints and choices. This process takes place in all realms of thought, including the arts.
2. Some of these mechanisms can be expressed in mathematical terms.
3. Some of them are physically realizable: the wheel, motors, bombs, digital computers, analogue computers, etc.
4. Certain mental mechanisms may correspond to certain mechanisms of nature.
5. Certain mechanizable aspects of artistic creation may be simulated by certain physical mechanisms or machines which exist or may be created.
6. It happens that computers can be useful in certain ways.

Here then is the theoretical point of departure for a utilization of electronic computers in musical composition.

We may further establish that the role of the living composer seems to have evolved, on the one hand, to one of inventing schemes (previously forms) and exploring the limits of these schemes, and on the other, to effecting the scientific synthesis of the new methods of construction and of sound emission. In a short while these methods must comprise all the ancient and modern means of musical instrument making, whether acoustic or electronic, with the help, for example, of digital-to-analogue converters; these have already been used in communication studies by N. Guttman, J. R. Pierce, and M. V. Mathews of Bell Telephone Laboratories in New Jersey. Now these explorations necessitate impressive mathematical, logical, physical, and psychological impedimenta, especially computers that accelerate the mental processes necessary for clearing the way for new fields by providing immediate experimental verifications at all stages of musical construction.

Music, by its very abstract nature, is the first of the arts to have attempted the conciliation of artistic creation with scientific thought. Its industrialization is inevitable and irreversible. Have we not already seen attempts to industrialize serial and popular music by the Parisian team of P. Barbaud, P. Blanchard, and Jeanine Charbonnier, as well as by the musicological research of Hiller and Isaacson at the University of Illinois?

In the preceding chapters we demonstrated some new areas of musical creation: Poisson, Markov processes, musical games, the thesis of the minimum of constraints, etc. They are all based on mathematics and especially on the theory of probability. They therefore lend themselves to being treated and

A figure?

Figure 1: Excerpt from the First Flow Chart of *Achorripsis*

explored by computers. The simplest and most meaningful scheme is one of minimum constraints in composition, as exemplified by *Achorripsis*.

Thanks to my friend Georges Boudouris of the C.N.R.S. I made the acquaintance of Jacques Barraud, Engineer of the Ecole des Mines, then director of the Ensemble Electroniques de Gestion de la Société des Petroles Shell-Berre, and François Genuys, agrégé in mathematics, and head of the Etudes Scientifiques Nouvelles at IBM-France. All three are scientists, yet they consented to attempt an experiment which seemed at first far-fetched—that of a marriage of music with one of the most powerful machines in the world.

In most human relations it is rarely pure logical persuasion which is important; usually the paramount consideration is material interest. Now in this case it was not logic, much less self-interest, that arranged the betrothal, but purely experiment for experiment's sake, or game for game's sake, that induced collaboration. Stochastically speaking, my venture should have encountered failure. Yet the doors were opened, and at the end of a year and a half of contacts and hard work “the most unusual event witnessed by the firm or by this musical season [in Paris]” took place on 24 May 1962 at the headquarters of IBM-France. It was a live concert presenting a work of stochastic instrumental music entitled *ST/10-1, 080262*, which had been calculated on the IBM-7090. It was brilliantly performed by the conductor C. Simonovic and his Ensemble de Musique Contemporaine de Paris. By its passage through the machine, this work made tangible a stochastic method of composition, that of the minimum of constraints and rules.

Position of the Problem

The first working phase was the drawing up of the flow chart, i.e., writing down clearly and in order the stages of the operations of the scheme of *Achorripsis*,¹ and adapting it to the machine structure. In the first chapter we set out the entire synthetic method of this minimal structure. Since the machine is an iterative apparatus and performs these iterations with extraordinary speed, the thesis had to be broken down into a sequential series of operations reiterated in loops. An excerpt from the first flow chart is shown in [Fig. V-1](#).

The statement of the thesis of *Achorripsis* receives its first machine-oriented interpretation in the following manner:

1. *The work consists of a succession of sequences or movements each a_i seconds long.* Their durations are totally independent (asymmetric) but have a fixed mean duration, which is introduced in the form of a parameter. These durations and their stochastic succession are given by the formula

$$P_{a_i} = ce^{-ca_i} da_i \quad (\text{See Appendix I.})$$

2. **Definition of the mean density of the sounds during a_i *.* During a sequence sounds are emitted from several sonic sources. If the total number of these sounds or points during a sequence is N_{a_i} , the mean density of this point cluster is N_{a_i}/a_i sounds/sec. In general, for a given instrumental ensemble this density has limits that depend on the number of instrumentalists, the nature of their instruments, and the technical difficulties of performance. For a large orchestra the upper limit is of the order of 150 sounds/sec. The lower limit ($V3$) is arbitrary and positive. We choose $(V3) = 0.11$ sounds/sec. Previous experiments led us to adopt a logarithmic progression for the density sensation with a number

¹See *Gravesaner Blätter*, nos. 11/12 (Mainz: Ars Viva Verlag, 1957).

Figure 2: *ST/10-1, 080262, Composition of the Orchestra*

between 2 and 3 as its base. We adopted $e = 2.71827$. Thus the densities are included between $(V3)e^0$ and $(V3)e^R$ sounds/sec., which we can draw on a line graduated logarithmically (base e).² As our purpose is total independence, we attribute to each of the sequences a_i calculated in ****1.**** a density represented by a point drawn at random from the portion of the line mentioned above. However a certain concern for continuity leads us to temper the independence of the densities among sequences a_i ; to this end we introduce a certain "memory" from sequence to sequence in the following manner:

Let a_{i-1} be a sequence of duration a_{i-1} , $(DA)_{i-1}$ its density, and a_i the next sequence with duration a_i and density $(DA)_i$. Density $(DA)_i$ will be given by the formula:

$$(DA)_i = (DA)_{i-1}e^{\pm x},$$

in which x is a segment of line drawn at random from a line segment s of length equal to $(R - 0)$. The probability of x is given by

$$P_x = \frac{2}{s} \left(1 - \frac{x}{s}\right) dx$$

and finally,

$$N_{a_i} = (DA)_i a_i.$$

3. *Composition Q of the orchestra during sequence a_i^* . First the instruments are divided into r classes of timbres, e.g., flutes and clarinets, oboes and bassoons, brasses, bowed strings, pizzicati, col legno strokes, glissandi, wood, skin, and metal percussion instruments, etc. (See the table for Atrées.)
<!-- Composition of the Orchestra for Atrées (ST/10-3,060962) -->

The composition of the orchestra is stochastically conceived, i.e., the distribution of the classes is not deterministic. Thus during a sequence of duration a_i it may happen that we have 80Under actual conditions the determining factor which would condition the composition of the orchestra is density. We therefore connect the orchestral composition with density by means of a special diagram. An example from *ST/10-1, 080262* is shown in Fig. V-2. <!-- Fig V-2. ST/10-1, 080262, Composition of the Orchestra -->

Fig. V-2 is expressed by the formula

$$Q_r = (n - x)(e_{n,r} - e_{n+1,r}) + e_{n,r}$$

in which r = the number of the class, $x = \log_e [(DA)_i / (V3)]$, $n = 0, 1, 2, \dots, R$, such that $n \leq x \leq n + 1$, and $e_{n,r}$ and $e_{n+1,r}$ are the probabilities of class r as a function of n .

It goes without saying that the composition of this table is a precise task of great complexity and delicacy. Once these preliminaries have been completed, we can define, one after the other, the N_{a_i} , sounds of sequence a_i .

4. *Definition of the moment of occurrence of the sound N within the sequence a_i^* . The mean density of the points or sounds to be distributed within a_i is $k = N_{a_i} / a_i$. The formula which gives the intervals separating the sound attacks is

$$P_t = ke^{-kt} dt.$$

5. *Attribution to the above sound of an instrument belonging to orchestra Q , which has already been calculated*. First class r is drawn at random with probability q_r from the orchestra ensemble

² $(V3)e^R$ must be equal to the upper limit, e.g., to 150 sounds/sec. in the case of a large orchestra.

calculated in **3.** (Consider an urn with balls of r colors in various proportions.) Then from within class r the number of the instrument is drawn according to the probability p_n given by an arbitrary table (urn with balls of n colors). Here also the distribution of instruments within a class is delicate and complex.

6. *Attribution of a pitch as a function of the instrument*. Taking as the zero point the lowest MAKE A FLATThe range s of each instrument is thus expressed by a natural number (distance). But the pitch h_u of a sound is expressed by a decimal number of which the whole number part is related to a note of the chromatic scale within the instrument's range.

Just as for the density in **2.**, we accept a certain memory of or dependence on the preceding pitch played by the same instrument, so that we have

$$h_u = h_{u-1} \pm z,$$

where z is given by the probability formula

$$P_z = \frac{2}{s} \left(1 - \frac{z}{s}\right) dz.$$

P_z is the probability of the interval z taken at random from the range s , and s is expressed as the difference between the highest and lowest pitches that can be played on the instrument.

7. *Attribution of a glissando speed if class r is characterized as a glissando*. The homogeneity hypotheses in Chap. I led us to the formula

$$f(v) = \frac{2}{a\sqrt{\pi}} e^{-v^2/a^2},$$

and by the transformation $v/a = u$ to its homologue:

$$T(u) = \frac{2}{\sqrt{\pi}} \int_0^u e^{-u^2} du,$$

for which there are tables. $f(v)$ is the probability of occurrence of the speed v (which is expressed in semitones/sec.); it has a parameter a , which is proportional to the standard deviation s ($a = s\sqrt{2}$).

a is defined as a function of the logarithm of the density of sequence a_i by: an inversely proportional function

$$a = \sqrt{\pi} \left(30 - \frac{20}{R} L[(DA_i)/(V3)]\right),$$

or a directly proportional function

$$a = \sqrt{\pi} \left(10 + \frac{20}{R} L[(DA_i)/(V3)]\right),$$

or a function independent of density

$$a = 17.7 + 35k,$$

where k is a random number between 0 and 1.

The constants of the preceding formulae derive from the limits of the speeds that string glissandi may take.

Thus for $(DA)_i = 145$ sounds/sec.

$$a = 53.2 \text{ semitones/sec.}$$

$$2s = 75 \text{ semitones/sec.}$$

and for for $(DA)_i = 0.13$ sounds/sec.

$$a = 17.7 \text{ semitones/sec.}$$

$$2s = 25 \text{ semitones/sec.}$$

8. *Attribution of a duration x to the sounds emitted*. To simplify we establish a mean duration for each instrument, which is independent of tessitura and nuance. Consequently we reserve the right to modify it when transcribing into traditional notation. The following is the list of constraints that we take into account for the establishment of duration x :

G , the maximum length of respiration or desired duration

$(DA)_i$, the the density of the sequence

q_r , the probability of class r

p_n , the probability of the instrument n

Then if we define z as a parameter of a sound's duration, z could be inversely proportional to the probability of the occurrence of the instrument, so that

$$z = \frac{1}{(DA)_i p_n q_r}.$$

z will be at its maximum when $(DA)_i p_n q_r$ is at its minimum, and in this case we could choose $z_{\max} = G$.

Instead of letting $z_{\max} = G$, we shall establish a logarithmic law so as to freeze the growth of z . This law applies for any given value of z .

$$z' = G \ln z / \ln z_{\max}$$

Since we admit a total independence, the distribution of the durations x will be Gaussian:

$$f(x) = \frac{1}{s\sqrt{2\pi}} e^{-(x-m)^2/(2s^2)},$$

where m is the arithmetic mean of the durations, s the standard deviation, and

$$m - 4.25s = 0$$

$$m + 4.25s = z'$$

the linear system which furnishes us with the constants m and s . By assuming $u = (x - m)/s\sqrt{2}$ we find the function $T(u)$, for which we consult the tables.

Finally, the duration x of the sound will be given by the relation

$$x = \pm us\sqrt{2} + m.$$

We do not take into account incompatibilities between instruments, for this would needlessly burden the machine's program and calculation.

9. *Attribution of dynamic forms to the sounds emitted*. We define four zones of mean intensities: *pp*, *p*, *f*, *ff*. Taken three at a time they yield $4^3 = 64$ permutations, of which 44 are different (an urn with 44 colors); for example, FIX <!-- Table of the 44 Intensity Forms Derived from 4 Mean Intensity Values, ppp, p, f, ff -->

10. The same operations are begun again for each sound of the cluster N_{a_i} . 11. Recalculations of the same sort are made for the other sequences.

An extract from the sequential statement was reproduced in Fig. V-1. Now we must proceed to the transcription into Fortran IV, a language "understood" by the machine (see Fig. V-3). <!-- Fig. V-3. Stochastic Music Rewritten in Fortran IV --> <!-- Fig. V-4. Provisional Results of One Phase of the Analysis --> <!-- Fig. V-5. Bars 1-5 of ST/10-1,080262 -->

It is not our purpose to describe the transformation of the flow chart into Fortran. However, it would be interesting to show an example of the adaptation of a mathematical expression to machine methods.

Let us consider the elementary law of probability (density function)

$$f(x) dx = ce^{-cx} dx.$$

How shall we proceed in order for the computer to give us lengths x with the probability $f(x) dx$? The machine can only draw random numbers y_0 with equiprobability between 0 and 1. We shall "modulate" this probability: Assume some length x_0 ; then we have

$$\text{prob}(0 \leq x \leq x_0) = \int_0^{x_0} f(x) dx = 1 - e^{-cx_0} = F(x_0),$$

where $F(x_0)$ is the distribution function of x . But

$$F(x_0) = \text{prob}(0 \leq y \leq y_0) = y_0$$

then

$$1 - e^{-cx_0} = y_0$$

and

$$x_0 = -\frac{\ln(1 - y_0)}{c}$$

for all $x_0 \geq 0$.

Once the program is transcribed into language that the machine's internal organization can assimilate, a process that can take several months, we can proceed to punching the cards and setting up certain tests. Short sections are run on the machine to detect errors of logic and orthography and to determine the values of the entry parameters, which are introduced in the form of variables. This is a very important phase, for it permits us to explore all parts of the program and determine the modalities of its operation. The final phase is the decoding of the results into traditional notation, unless an automatic transcriber is available.

Conclusions

A large number of compositions of the same kind as *ST/10-1, 080262* is possible for a large number of orchestral combinations. Other works have already been written: *ST/48-1, 240162*, for large orchestra, commissioned by RTF (France III); *Atrées* for ten soloists; and *Morisma-Amorisima*, for four soloists.

Although this program gives a satisfactory solution to the minimal structure, it is, however, necessary to jump to the stage of pure composition by coupling a digital-to-analogue converter to the computer. The numerical calculations would then be changed into sound, whose internal organization had been conceived beforehand. At this point one could bring to fruition and generalize the concepts described in the preceding chapters.

The following are several of the advantages of using electronic computers in musical composition:

1. The long laborious calculation made by hand is reduced to nothing. The speed of a machine such as the IBM-7090 is tremendous—of the order of 500,000 elementary operations/sec.

2. Freed from tedious calculations the composer is able to devote himself to the general problems that the new musical form poses and to explore the nooks and crannies of this form while modifying the values of the input data. For example, he may test all instrumental combinations from soloists to chamber orchestras, to large orchestras. With the aid of electronic computers the composer becomes a sort of pilot: he presses the buttons, introduces coordinates, and supervises the controls of a cosmic vessel sailing in the space of sound, across sonic constellations and galaxies that he could formerly glimpse only as a distant dream. Now he can explore them at his ease, seated in an armchair. 3. The program, i.e., the list of sequential operations that constitute the new musical form, is an objective manifestation of this form. The program may consequently be dispatched to any point on the earth that possesses computers of the appropriate type, and may be exploited by any composer pilot. 4. Because of certain uncertainties introduced in the program, the composer-pilot can instill his own personality in the sonic result he obtains.

C	PROGRAM FREE STOCHASTIC MUSIC (FORTRAN IV)	XEN	6
C		XEN	7
C	GLOSSARY OF THE PRINCIPAL ABBREVIATIONS	XEN	8
C			
C	A - DURATION OF EACH SEQUENCE IN SECONDS	XEN	9
C	A10,A20,A17,A35,A30 - NUMBERS FOR GLISSANDO CALCULATION	XEN	10
C	ALEA - PARAMETER USED TO ALTER THE RESULT OF A SECOND RUN WITH THE	XEN	11
C	SAME INPUT DATA	XEN	12
C	ALFA(3) - THREE EXPRESSIONS ENTERING INTO THE THREE SPEED VALUES	XEN	13
C	OF THE SLIDING TONES (GLISSANDI)	XEN	14
C	ALIM - MAXIMUM LIMIT OF SEQUENCE DURATION A	XEN	15
C	(AMAX(I),I=1,KTR) TABLE OF AN EXPRESSION ENTERING INTO THE	XEN	16
C	CALCULATION OF THE NOTE LENGTH IN PART 8	XEN	17
C	BF - DYNAMIC FORM NUMBER, THE LIST IS ESTABLISHED INDEPENDENTLY	XEN	18
C	OF THIS PROGRAM AND IS SUBJECT TO MODIFICATION	XEN	19
C	DELTA - THE RECIPROCAL OF THE MEAN DENSITY OF SOUND EVENTS DURING	XEN	20
C	A SEQUENCE OF DURATION A	XEN	21
C	(E(I,J),I=1,KTR,J=1,KTE) - PROBABILITIES OF THE KTR TIMBRE CLASSES	XEN	22
C	INTRODUCED AS INPUT DATA, DEPENDING ON THE CLASS NUMBER I=KR	XEN	23
C	AND ON THE POWER J=U OBTAINED FROM $V3 \cdot \text{EXP}(U) = DA$	XEN	24
C	EPSI - EPSILON FOR ACCURACY IN CALCULATING PN AND E(I,J), WHICH	XEN	25
C	IT IS ADVISABLE TO RETAIN.	XEN	26
C	(GN(I,J),I=1,KTR,J=1,KTS) - TABLE OF THE GIVEN LENGTH OF BREATH	XEN	27
C	FOR EACH INSTRUMENT, DEPENDING ON CLASS I AND INSTRUMENT J	XEN	28
C	GTNA - GREATEST NUMBER OF NOTES IN THE SEQUENCE OF DURATION A	XEN	29
C	GTNS - GREATEST NUMBER OF NOTES IN KW LOOPS	XEN	30
C	(HAMIN(I,J),HAMAX(I,J),HBMIN(I,J),HBMAX(I,J),I=1,KTR,J=1,KTS)	XEN	31
C	TABLE OF INSTRUMENT COMPASS LIMITS, DEPENDING ON TIMBRE CLASS I	XEN	32
C	AND INSTRUMENT J. TEST INSTRUCTION 480 IN PART 6 DETERMINES	XEN	33
C	WHETHER THE HA OR THE HB TABLE IS FOLLOWED. THE NUMBER 7 IS	XEN	34
C	ARBITRARY.	XEN	35
C	JW - ORDINAL NUMBER OF THE SEQUENCE COMPUTED.	XEN	36
C	KNL - NUMBER OF LINES PER PAGE OF THE PRINTED RESULT.KNL=50	XEN	37
C	KR1 - NUMBER IN THE CLASS KR=1 USED FOR PERCUSSION OR INSTRUMENTS	XEN	38
C	WITHOUT A DEFINITE PITCH.	XEN	39
C	KTE - POWER OF THE EXPONENTIAL COEFFICIENT E SUCH THAT	XEN	40
C	$DA(\text{MAX}) = V3 \cdot (E^{**}(KTE-1))$	XEN	41
C	KTR - NUMBER OF TIMBRE CLASSES	XEN	42
C	KW - MAXIMUM NUMBER OF JW	XEN	43
C	KTEST1,TAV1,ETC. - EXPRESSIONS USEFUL IN CALCULATING HOW LONG THE	XEN	44
C	VARIOUS PARTS OF THE PROGRAM WILL RUN.	XEN	45

C	KT1 - ZERO IF THE PROGRAM IS BEING RUN, NONZERO DURING DEBUGGING	XEN	46
C	KT2 - NUMBER OF LOOPS, EQUAL TO 15 BY ARBITRARY DEFINITION.	XEN	47
C	(MODI(IX8),IX8=7,1) AUXILIARY FUNCTION TO INTERPOLATE VALUES IN	XEN	48
C	THE TETA(256) TABLE (SEE PART 7)	XEN	49
C	NA - NUMBER OF SOUNDS CALCULATED FOR THE SEQUENCE A (NA=DA*A)	XEN	50
C	(NT(I),I=1,KTR) NUMBER OF INSTRUMENTS ALLOCATED TO EACH OF THE	XEN	51
C	KTR TIMBRE CLASSES.	XEN	52
C	(PN(I,J),I=1,KTR,J=1,KTS),(KTS=NT(I),I=1,KTR) TABLE OF PROBABILITY	XEN	53
C	OF EACH INSTRUMENT OF THE CLASS I.	XEN	54
C	(Q(I),I=1,KTR) PROBABILITIES OF THE KTR TIMBRE CLASSES, CONSIDERED	XEN	55
C	AS LINEAR FUNCTIONS OF THE DENSITY DA.	XEN	56
C	(S(I),I=1,KTR) SUM OF THE SUCCESSIVE Q(I) PROBABILITIES, USED TO	XEN	57
C	CHOOSE THE CLASS KR BY COMPARING IT TO A RANDOM NUMBER X1 (SEE	XEN	58
C	PART 3, LOOP 380 AND PART 5, LOOP 430).	XEN	59
C	SINA - SUM OF THE COMPUTED NOTES IN THE NEW CLOUDS NA, ALWAYS LESS	XEN	60
C	THAN GTNS (SEE TEST IN PART 10).	XEN	61
C	SQPI - SQUARE ROOT OF PI (3.14159...)	XEN	62
C	TA - SOUND ATTACK TIME ABCISSA.	XEN	63
C	TETA(256) - TABLE OF THE 256 VALUES OF THE INTEGRAL OF THE NORMAL	XEN	64
C	DISTRIBUTION CURVE WHICH IS USEFUL IN CALCULATING GLISSANDO SPEED	XEN	65
C	AND SOUND EVENT DURATION.	XEN	66
C	VIGL - GLISSANDO SPEED (VITESSE GLISSANDO), WHICH CAN VARY AS, BE	XEN	67
C	INDEPENDENT OF, OR VARY INVERSELY AS THE DENSITY OF THE SEQUENCE,	XEN	68
C	THE ACTUAL MODE OF VARIATION EMPLOYED REMAINING THE SAME FOR THE	XEN	69
C	ENTIRE SEQUENCE (SEE PART 7).	XEN	70
C	VITLIM - MAXIMUM LIMITING GLISSANDO SPEED (IN SEMITONES/SEC),	XEN	71
C	SUBJECT TO MODIFICATION.	XEN	72
C	V3 - MINIMUM CLOUD DENSITY DA	XEN	73
C	(Z1(I),Z2(I),I=1,8) TABLE COMPLEMENTARY TO THE TETA TABLE.	XEN	74
C		XEN	75
C		XEN	76
C	READ CONSTANTS AND TABLES	XEN	78
C		XEN	77
	DIMENSION Q(12),S(12),E(12,12),PN(12,50),SPN(12,50),NT(12),	XEN	79
	1HAMIN(12,50),HAMAX(12,50),HBMIN(12,50),HBMAX(12,50),GN(12,50),H(12	XEN	80
	2,50),TETA(256),VIGL(3),MODI(7),Z1(8),Z2(8),ALFA(3),AMAX(12)	XEN	81
C		XEN	82
C		XEN	83
C		XEN	84
	I=1	XEN	85
	DO 10 IX=1,7	XEN	86
	IX8=8-IX	XEN	87
	MODI(IX8)=I	XEN	88
10	I=I+1	XEN	89
C		XEN	90
	READ 20,(TETA(I),I=1,256)	XEN	91
20	FORMAT(12F6.6)	XEN	92
	READ 30,(Z1(I),Z2(I),I=1,8)	XEN	93
30	FORMAT(6(F3.2,F9.8)/F3.2,F9.8,E6.2,F9.8)	XEN	94
	PRINT 40,TETA,Z1,Z2	XEN	95
40	FORMAT('1THE TETA TABLE= ',/,21(12F10.6,/),4F10.6,/////,	XEN	96
	1'THE Z1 TABLE= ',/,7F6.2,E12.3,///, 'THE Z2 TABLE= ',/,8F14.8,/,	XEN	97
	21H1)	XEN	98
	READ 50,DELTA,V3,A10,A20,A17,A30,A35,BF,SQPI,EPSI,VITLIM,ALEA,	AXEN	99
	1LIM	XEN	100
50	FORMAT(F3.0,F3.3,5F3.1,F2.0,F8.7,F8.8,F4.2,F8.8,F5.2)	XEN	114
	READ 60,KT1,KT2,KW,KNL,KTR,KTE,KR1,GTNA,GTNS,(NT(I),I=1,KTR)	XEN	115
60	FORMAT(5I3,2I2,2F6.0,12I2)	XEN	126
	PRINT 70,DELTA,V3,A10,A20,A17,A30,A35,BF,SQPI,EPSI,VITLIM,ALEA,	AXEN	127

```

1LIM,KT1,KT2,KW,KNL,KTR,KTE,KR1,GTNA,GTNS,(I,NT(I),I=1,KTR) XEN 128
70  FORMAT('1DELTA= ',F4.0/, 'V3= ',F6.3/, 'A10= ',F4.1/, XEN 129
1  'A20= ',F4.1/, 'A17= ',F4.1/, 'A30= ',F4.1/, 'A35= ',F4.1,XEN 130
2/, 'BF= ',F3.0/, 'SQPI= ',F11.8/, 'EPSI= ',F12.8/, 'VITLIM= ',XEN 131
3,F5.2/, 'ALEA= ',F12.8/, 'ALIM= ',F6.2/, 'KT1= ',I3/, XEN 132
4'KT2= ',I3/, 'KW= ',I3/, 'KNL= ',I3/, 'KTR= ',I3/, XEN 133
5'KTE= ',I2/, 'KR1= ',I2/, 'GTNA= ',F7.0/, 'GTNS= ',F7.0, XEN 134
6/,12('INCLASS',I2, 'THEREARE',I2, 'INSTRUMENTS.',/)) XEN 135
  READ 80,KTEST3,KTEST1,KTEST2 XEN 136
80  FORMAT(5I3) XEN 141
  PRINT 90,KTEST3,KTEST1,KTEST2 XEN 142
90  FORMAT('KTEST3= ',I3/, 'KTEST1= ',I3/, 'KTEST2= ',I3) XEN 143
C XEN 144
  IF(KTEST3.NE.0) PRINT 830 XEN 145
  R=KTE-1 XEN 146
  A10=A10*SQPI XEN 147
  A20=A20*SQPI/R XEN 148
  A30=A30*SQPI XEN 149
C  IF ALEA IS NON-ZERO, THE RANDOM NUMBER IS GENERATED FROM THE TIME XEN 150
C  WHEN THE FOLLOWING INSTRUCTION IS EXECUTED. IF ALEA IS NON-ZERO XEN 151
C  EACH RUN OF THIS PROGRAM WILL PRODUCE DIFFERENT OUTPUT DATA. XEN 152
  IF(ALEA.NE.0.0) CALL RANFSET(TIMEF(1)) XEN 153
  PRINT 830 XEN 154
  DO 130 I=1,KTR XEN 155
  Y=0.0 XEN 156
  KTS=NT(I) XEN 157
  READ 100,(HAMIN(I,J),HAMAX(I,J),HBMIN(I,J),HBMAX(I,J),GN(I,J), XEN 158
1PN(I,J),J=1,KTS) XEN 159
100  FORMAT(5(5F2.0,F3.3)) XEN 160
  PRINT 110,I,(J,HAMIN(I,J),HAMAX(I,J),HBMIN(I,J),HBMAX(I,J),GN(I,J) XEN 161
1,PN(I,J),J=1,KTS) XEN 162
110  FORMAT(////, 'INCLASSNUMBER',I2/,('FORINSTRUMENTNO.',I2, XEN 163
1'HAMIN= ',F3.0, 'HAMAX= ',F3.0, 'HBMIN= ',F3.0, 'HBMAX= ', XEN 164
2F3.0, 'GN= ',F3.0, 'ANDPN= ',F6.3)) XEN 165
  DO 120 J=1,KTS XEN 166
  Y=Y+PN(I,J) XEN 167
120  SPN(I,J)=Y XEN 168
130  IF(ABSF(Y-1.0).GE.EPSI) CALL EXIT XEN 169
C XEN 170
  DO 150 I=1,KTR XEN 171
  READ 140,(E(I,J),J=1,KTE) XEN 172
140  FORMAT(12F2.2) XEN 173
150  PRINT 160,I,(J,E(I,J),J=1,KTE) XEN 174
160  FORMAT(////////, 'CLASSNUMBER',I2/,('INDENSITYLEVEL',I2, XEN 175
1'HASAUPPROBABILITYOF',F6.2)) XEN 176
  DO 180 J=1,KTE XEN 177
  Y=0.0 XEN 178
  DO 170 I=1,KTR XEN 179
170  Y=Y+E(I,J) XEN 180
180  IF(ABSF(Y-1.0).GE.EPSI) CALL EXIT XEN 181
  DO 200 I=1,KTR XEN 182
  AMAX(I)=1.0/E(I,1) XEN 183
  DO 200 J=2,KTE XEN 184
  AJ=J-1 XEN 185
  AX=1.0/(E(I,J)*EXPF(AJ)) XEN 186
  IF (KT1.NE.0) PRINT 190,AX XEN 187
190  FORMAT(1H ,9E12.8) XEN 188
200  IF(AX.GT.AMAX(I)) AMAX(I)=AX XEN 189
  IF(KT1.NE.0) WRITE(2,210) AMAX XEN 190

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210	FORMAT(6E15.8)	XEN	191
C		XEN	192
	JW=1	XEN	193
	SINA=0.0	XEN	194
	IF(KTEST1.NE.0) TAV1=TIMEF(1)	XEN	195
220	NLINE=50	XEN	196
C		XEN	197
C	PARTS 1 AND 2, DEFINE SEQUENCE A SECONDS AND CLOUD NA DURING A	XEN	198
C		XEN	199
	KNA=0	XEN	200
	K1=0	XEN	201
230	X1=RANF(-1)	XEN	202
	A=-DELTA * LOGF(X1)	XEN	203
	IF(A.LE.ALIM) GO TO 250	XEN	204
	IF(K1.GE.KT2) GO TO 240	XEN	205
	K1=K1+1	XEN	206
	GO TO 230	XEN	207
240	A=ALIM/2.0	XEN	208
	X1=0.0	XEN	209
250	K2=0	XEN	210
260	X2=RANF(-1)	XEN	211
	IF(JW.GT.1) GO TO 280	XEN	212
270	UX=R*X2	XEN	213
	GO TO 310	XEN	214
280	IF(RANF(-1).GE.0.5) GO TO 290	XEN	215
	UX=UPR + R * (1.0-SQRTF(X2))	XEN	216
	GO TO 300	XEN	217
290	UX=UPR - R * (1.0-SQRTF(X2))	XEN	218
300	IF((UX.GE.0.0).AND.(UX.LE.R)) GO TO 310	XEN	219
	IF(K2.GE.KT2) GO TO 270	XEN	220
	K2=K2+1	XEN	221
	GO TO 260	XEN	222
310	U=UX	XEN	223
	DA=V3*EXPF(U)	XEN	224
	NA=XINTF(A * DA + 0.5) + 1	XEN	225
	IF (GTNA.GT.FLOATF(NA)) GO TO 330	XEN	226
	IF (KNA.GE.KT2) GO TO 320	XEN	227
	KNA=KNA+1	XEN	228
	GO TO 230	XEN	229
320	A=DELTA	XEN	230
	GO TO 260	XEN	231
330	UPR=U	XEN	232
	IF (KT1.EQ.0) GO TO 360	XEN	233
	PRINT 340, JW, KNA, K1, K2, X1, X2, A, DA, NA	XEN	234
340	FORMAT(1H1,4I8,3X,4E18.8,3X,I8)	XEN	235
	NA=KT1	XEN	236
	IF (KTEST3.NE.0) PRINT 350, JW, NA, A	XEN	237
350	FORMAT(1H0,2I9,F10.2)	XEN	238
C		XEN	239
C	PART 3. DEFINE CONSTITUTION OF ORCHESTRA DURING SEQUENCE A	XEN	240
C		XEN	241
360	SINA=SINA + FLOATF(NA)	XEN	242
	XLOGDA=U	XEN	243
	XALOG=A20 *XLOGDA	XEN	244
	M=XINTF(XLOGDA)	XEN	245
	IF ((M+2).GT.KTE) M=KTE-2	XEN	246
	SR=0.0	XEN	247
	M1=M+1	XEN	248
	M2=M+2	XEN	249

DO 380 I=1,KTR	XEN	250
ALFX=E(I,M1)	XEN	251
BETA=E(I,M2)	XEN	252
XM=M	XEN	253
QR=(XLOGDA-XM) * (BETA-ALFX) + ALFX	XEN	254
IF (KT1.NE.0) PRINT 370,XM,ALFX,BETA	XEN	255
370 FORMAT(1H ,3F20.8)	XEN	256
Q(I)=QR	XEN	257
SR=SR+QR	XEN	258
380 S(I)=SR	XEN	259
IF (KT1.NE.0) PRINT 390,(Q(I),I=1,KTR),(S(I),I=1,KTR)	XEN	260
390 FORMAT(1H ,12F9.4)	XEN	261
C	XEN	262
C PART 4,DEFINE INSTANT TA OF EACH POINT IN SEQUENCE A	XEN	263
C	XEN	264
IF (KTEST2.NE.0) TAV2=TIMEF(1)	XEN	265
N=1	XEN	266
T=0.0	XEN	267
TA=0.0	XEN	268
GO TO 410	XEN	269
400 N=N+1	XEN	270
X=RANF(-1)	XEN	271
T=-LOGF(X)/DA	XEN	272
TA=TA+T	XEN	273
410 IF (KT1.NE.0) PRINT 420,N,X,T,TA	XEN	274
420 FORMAT(//,I8,3E20.8)	XEN	275
C	XEN	276
C PART 5,DEFINE CLASS AND INSTRUMENT NUMBER TO EACH POINT OF A	XEN	277
C	XEN	278
X1=RANF(-1)	XEN	279
DO 430 I=1,KTR	XEN	280
430 IF (X1.LE.S(I)) GO TO 440	XEN	281
I=KTR	XEN	282
440 KTS=NT(I)	XEN	283
KR=I	XEN	284
X2=RANF(-1)	XEN	285
DO 450 J=1,KTS	XEN	286
SPIEN=SPN(KR,J)	XEN	287
INSTRM=J	XEN	288
450 IF (X2.LE.SPIEN) GO TO 460	XEN	289
INSTRM=KTS	XEN	290
460 PIEN=PN(KR,INSTRM)	XEN	291
IF (KT1.NE.0) PRINT 470,X1,S(KR),KR,X2,SPIEN,INSTRM	XEN	292
470 FORMAT(1H ,2E20.8,I6,2E20.8,I6)	XEN	293
C	XEN	294
C PART 6,DEFINE PITCH HN FOR EACH POINT OF SEQUENCE A	XEN	295
C	XEN	296
IF (KR.GT.1) GO TO 480	XEN	297
IF (INSTRM.GE.KR1) GO TO 490	XEN	298
HX=0.0	XEN	299
GO TO 560	XEN	300
480 IF (KR.LT.7) GO TO 490	XEN	301
HSUP=HBMAX(KR,INSTRM)	XEN	302
HINF=HBMIN(KR,INSTRM)	XEN	303
GO TO 500	XEN	304
490 HSUP=HAMAX(KR,INSTRM)	XEN	305
HINF=HAMIN(KR,INSTRM)	XEN	306
500 HM=HSUP-HINF	XEN	307
HPR=H(KR,INSTRM)	XEN	308

K=0	XEN	309
IF (HPR.LE.0.0) GO TO 520	XEN	310
510 X=RANF(-1)	XEN	311
IF (N.GT.1) GO TO 530	XEN	312
520 HX=HINF+HM*X RANF(-1)	XEN	313
GO TO 560	XEN	314
530 IF(RANF(-1).GE.0.5) GO TO 540	XEN	315
HX=HPR+HM * (1.0-SQRTF(X))	XEN	316
GO TO 550	XEN	317
540 HX=HPR-HM * (1.0-SQRTF(X))	XEN	318
550 IF((HX.GE.HINF).AND.(HX.LE.HSUP)) GO TO 560	XEN	319
IF (K.GE.KT2) GO TO 520	XEN	320
K=K+1	XEN	321
GO TO 510	XEN	322
560 H(KR, INSTRM)=HX	XEN	323
IF (KT1.NE.0) PRINT 570,K,X,HX	XEN	324
570 FORMAT(1H ,I6,2E20.8)	XEN	325
C	XEN	326
C PART 7,DEFINE SPEED VIGL TO EACH POINT OF A	XEN	327
C	XEN	328
IF (KR.EQ.5) GO TO 580	XEN	329
VIGL(1)=0.0	XEN	330
VIGL(2)=0.0	XEN	331
VIGL(3)=0.0	XEN	332
X1=0.0	XEN	333
X2=0.0	XEN	334
XLAMBDA=0.0	XEN	335
GO TO 740	XEN	336
580 KX=1	XEN	337
590 X1=RANF(-1)	XEN	338
IF (X1-0.9997) 600,650,680	XEN	339
600 I=128	XEN	340
DO 630 IX=1,7	XEN	341
IF(TETA(I)-X1) 610,640,620	XEN	342
610 I=I+MODI(IX)	XEN	343
GO TO 630	XEN	344
620 I=I-MODI(IX)	XEN	345
630 CONTINUE	XEN	346
IF(TETA(I)-X1) 670,640,660	XEN	347
640 XLAMBDA=FLOATF(I-1)/100.0	XEN	348
GO TO (720,760), KX	XEN	349
650 XLAMBDA=2.55	XEN	350
GO TO (720,760), KX	XEN	351
660 I=I-1	XEN	352
670 TX1=TETA(I)	XEN	353
XLAMBDA=(FLOATF(I-1)+(X1-TX1)/(TETA(I+1)-TX1))/100.0	XEN	354
GO TO (720,760), KX	XEN	355
680 DO 690 I=2,7	XEN	356
TX1=Z2(I)	XEN	357
IF(X1-TX1) 700,710,690	XEN	358
690 CONTINUE	XEN	359
I=8	XEN	360
TX1=1.0	XEN	361
700 TX2=Z1(I)	XEN	362
XLAMBDA=TX2-((TX1-X1)/(TX1-Z2(I-1)))*(TX2-Z1(I-1))	XEN	363
GO TO (720,760), KX	XEN	364
710 XLAMBDA=Z1(I)	XEN	365
GO TO(720,760), KX	XEN	366
720 ALFA(1)=A10+XALOG	XEN	367

ALFA(3)=A30-XALOG	XEN	368
X2=RANF(-1)	XEN	369
ALFA(2)=A17+A35*X2	XEN	370
DO 730 I=1,3	XEN	371
VIGL(I)=INTF(ALFA(I)*XLAMBDA+0.5)	XEN	372
IF (VIGL(I).LT.0.0) VIGL(I)=-VIGL(I)	XEN	373
IF (VIGL(I).GT.VITLIM) VIGL(I)=VITLIM	XEN	374
730 IF (RANF(-1).LT.0.5) VIGL(I)=-VIGL(I)	XEN	375
740 IF(KT1.NE.0) PRINT 750,X1,X2,XLAMBDA,VIGL	XEN	376
750 FORMAT(1H ,6E19.8)	XEN	377
C	XEN	378
C PART 8,DEFINE DURATION FOR EACH POINT OF A	XEN	379
C	XEN	380
IF ((KR.EQ.7).OR.(KR.EQ.8)) GO TO 780	XEN	381
ZMAX=AMAX(KR)/(V3*PIEN)	XEN	382
G=GN(KR, INSTRM)	XEN	383
RO=G/LOGF(ZMAX)	XEN	384
QPND=1.0/(Q(KR)*PIEN*DA)	XEN	385
GE=ABSF(RO*LOGF(QPND))	XEN	386
XMU=GE/2.0	XEN	387
SIGMA=GE/4.0	XEN	388
KX=2	XEN	389
GO TO 590	XEN	390
760 TAU=SIGMA*XLAMBDA*1.4142	XEN	391
X2=RANF(-1)	XEN	392
IF (X2.GE.0.5) GO TO 770	XEN	393
XDUR=XMU+TAU	XEN	394
GO TO 790	XEN	395
770 XDUR=XMU-TAU	XEN	396
IF (XDUR.GE.0.0) GO TO 790	XEN	397
780 XDUR=0.0	XEN	398
790 IF(KT1.NE.0)PRINT 800,ZMAX, XMU, SIGMA, X1, XLAMBDA, X2, XDUR	XEN	399
800 FORMAT(1H ,5E15.8,E11.4,E15.8)	XEN	400
C	XEN	401
C PART 9,DEFINE INTENSITY FORM TO EACH POINT OF A	XEN	402
C	XEN	403
IFORM=XINTF(RANF(-1)*BF+0.5)	XEN	404
IF (KT1.EQ.0) GO TO 840	XEN	405
IF (NLINE.LT.KNL) GO TO 810	XEN	406
IF (NLINE.EQ.KNL) GO TO 820	XEN	407
NLINE=1	XEN	408
GO TO 900	XEN	409
810 NLINE=NLINE+1	XEN	410
GO TO 900	XEN	411
820 PRINT 830	XEN	412
830 FORMAT(1H1)	XEN	413
NLINE=0	XEN	414
GO TO 900	XEN	415
840 IF (NLINE.GE.KNL) GO TO 850	XEN	416
NLINE=NLINE+1	XEN	417
GO TO 880	XEN	418
850 PRINT 860,JW,A,NA,(Q(I),I=1,KTR)	XEN	419
860 FORMAT('1□□JW=',I3,4X,'A=',F8.2,4X,'NA=',I6,4X,'Q(I)=',12(F4.2,'/'	XEN	420
1),//)	XEN	421
PRINT 870	XEN	422
870 FORMAT(6X,'N',8X,'START',5X,'CLASS',4X,'INSTRM',4X,'PITCH',6X,	XEN	423
1'GLISS1',4X,'GLISS2',4X,'GLISS3',8X,'DURATION',5X,'DYNAM')	XEN	424
NLINE=1	XEN	425
880 PRINT 890,N,TA,KR, INSTRM, HX, (VIGL(I),I=1,3), XDUR, IFORM	XEN	426

890	FORMAT(1H ,I7,F12.2,I9,I8,F11.1,F13.1,2F10.1,F14.2,I11)	XEN	427
C		XEN	428
C	PART 10,REPEAT SAME DEFINITIONS FOR ALL POINTS OF A	XEN	429
C		XEN	430
900	IF (N.LT.NA) GO TO 400	XEN	431
C		XEN	432
C	PART 11, REPEAT SEQUENCES A	XEN	433
C		XEN	434
	IF (KTEST2.EQ.0) GO TO 910	XEN	435
	TAP2=TIMEF(1)-TAV2	XEN	436
	TAP2=TAP2/FLOATF(NA)	XEN	437
	PRINT 750,TAP2	XEN	438
C		XEN	439
910	IF (JW.GE.KW) GO TO 930	XEN	440
920	JW=JW+1	XEN	441
	IF (GTNS.GT.SINA) GO TO 220	XEN	442
930	IF (KTEST1.EQ.0) CALL EXIT	XEN	443
940	TAP1=TIMEF(-1)-TAV1	XEN	444
	TAP1=TAP1/FLOATF(KW)	XEN	445
	PRINT 750,TAP1	XEN	446
C		XEN	447
	END	XEN	448
C			
C	DATA FOR ATREES (ST/10-3, 060962)		
C			
	000000011300022600033900045100056400067600078900090100101300112500123600		
	134800145900156900168000179000190000200900211800222700233500244300255000		
	265700276300286900297400307900318300328600338900349100359300369400379400		
	389300399200409000418700428400438000447500456900466200475500484700493700		
	502700511700520500529200537900546500554900563300571600579800587900595900		
	603900611700619400627000634600642000649400656600663800670800677800684700		
	691400698100704700711200717500723800730000736100742100748000753800759500		
	765100770700776100781400786700791800796900801900806800811600816300820900		
	825400829900834200838500842700846800850800854800858600862400866100869800		
	87330087680088020088350088680089000089310089610089910090200904800907600		
	910300913000915500918100920500922900925200927500929700931900934000936100		
	938100940000941900943800945700947300949000950700952300953800955400956900		
	958300959700961100962400963700964900966100967300968400969500970600971600		
	972600973600974500975500976300977200978000978800979600980400981100981800		
	982500983200983800984400985000985600986100986700987200987700988200988600		
	989100989500989900990300990700991100991500991800992200992500992800993100		
	993400993700993900994200994400994700994900995100995300995500995700995900		
	996100996300996400996600996700996900997000997200997300997400997500997600		
	997700997900997960998050998140998230998320998400998480998550998620998680		
	998740998800998850998910998960999010999060999100999140999180999230999270		
	999300999340999370999400999440999470999500999530999560999580999600999630		
	999650999670999690999700		
	255099970000263099980000275099990000313099999000346099999900377099999990		
	406099999999100E30100000000		
	04005010020017730035563177245390100000071001000000012000		
	0000150500500120720001600025000120101020309020201010202		
	0000000000		
	01010000100700101000010090010100001012001010000101100101000010090		
	01010000101200101000010080010100001008001010000101200101000010080		
	01010000150200101000020020		
	1755000010999		
	3975000015999		
	29710000206001754000010400		
	348500001540015630000154001953000010200		

Figure 3: *ST/10-1, 080262*, Composition of the Orchestra

39750000151502971000010090175400000709017550000100903363000010090
1953000010070101300001020034850000152001563000015020
00003467005000000154800500
00003467005000000154800500
0000326810999
0000336310999
00001953108000000101307200
00003487155000000157215500
25080408011309
08071602010110
03030420010110
02050325010112
03350315011505
02100302103907
02020203150207
02020202410207
03090317041609
03132003200509
02052801030409
45011202020106

Appendix I

Two Laws of Continuous Probability

First Law

Second Law